

Key Sections, Elevations & Material Palette

The canopy section solved against the shadow geometry

This is the drawing the whole scheme rests on: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m louvre on the southern face. Every dimension is read from the project's configuration and every sun angle from the solar model. None of it is drawn by eye.

Upload slot 04 — Key Sections, Elevations & Material Palette

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

A flat canopy fails in a specific and predictable way. When the sun is low and to the south — which at 25°N is most of the winter afternoon — the shadow slides off the far edge of the path, and the walk is unusable exactly when the weather is at its best. An earlier version of this project claimed 99.2% annual shade on such a canopy. It was withdrawn.

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N

June solstice noon · 87.3°

December solstice noon · 41.3°

Al Falaj — 0.9 m channel, set on the canopy's drip line so it is shaded all day. An open channel here evaporates.

NORTH (concave — the cool side)

1.75 m

4.3 m to the shading plane

3 m southern louvre — the piece that buys the winter. A mashrabiya's logic: a deep screen on the face the sun comes from.

Neem · 10 m crown at maturity

Ghaf · 12 m crown at maturity

7 m walk

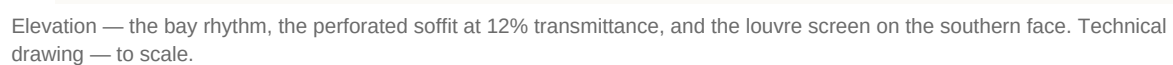
18 m gridshell — overhangs 5.5 m each side

SOUTH (convex)

The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.

Source: src/config.py CRESCENT; sun angles NREL SPA via pvliv · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

2. Elevation and structural rhythm



2

3. Why the tree avenue exists

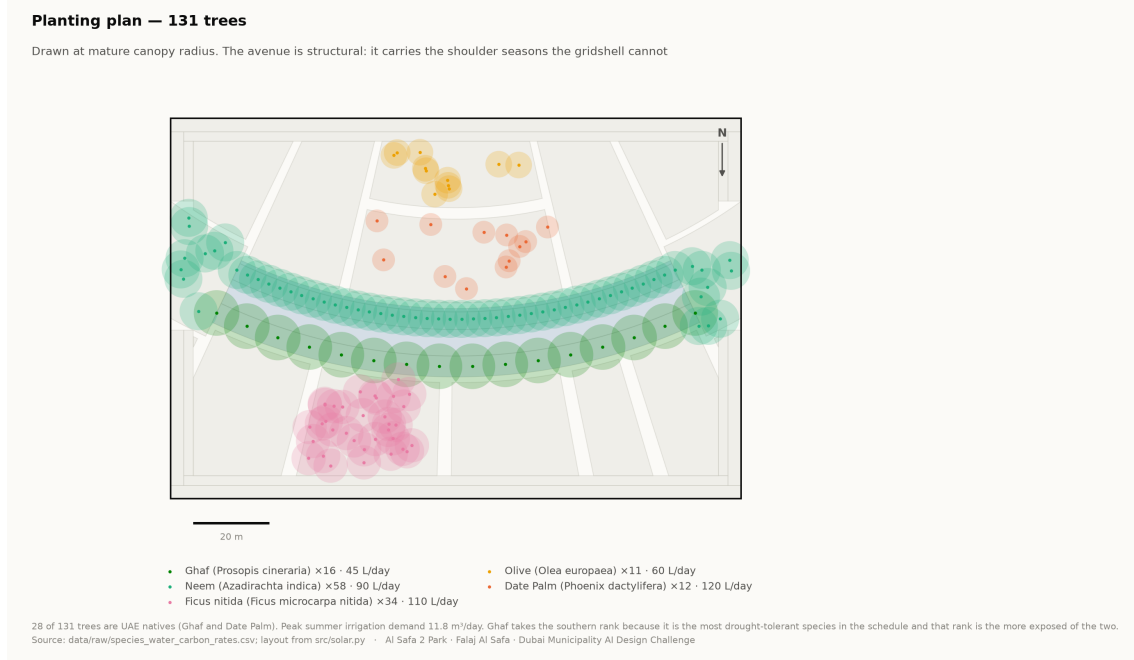
Shadow length is height divided by the tangent of solar elevation. At Dubai's summer noon a 6 m tree casts a 0.19 m shadow; at 20° elevation the same tree casts 16.5 m. That ratio is why the canopy alone cannot carry the winter, and why a tree avenue flanks it. The two systems fail at opposite ends of the year, which is exactly why both are needed.

4. Material and landscape palette

- **Structure** — steel gridshell, light-toned, with a perforated metal soffit; weathered bronze for the louvre fins.
- **Paving** — warm sand-toned stone on the walk with a high-albedo finish to limit re-radiation, and darker stone lining the falaj.
- **Earth** — Al Kathib is planted earth, not a wall. Cut and fill are balanced on site where possible.
- **Planting** — five desert species only. Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.

Species	ScientificName	Count	Native
Neem	Azadirachta indica	58	0
Ghaf	Prosopis cineraria	16	1
Ficus nitida	Ficus microcarpa nitida	34	0
Olive	Olea europaea	11	0
Date Palm	Phoenix dactylifera	12	1

131 trees. Source: data/raw/species_water_carbon_rates.csv.



Planting plan — 131 trees drawn at mature canopy radius. Technical drawing — to scale.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.